AI-ASSISTED WEARABLE MOBILITY AID FOR VISUALLY IMPAIRED

MINI PROJECT REPORT

submitted by

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in partial fulfillment of the requirements

for the award of the degree of

BACHELOR OF TECHNOLOGY



DEPARTMENT OF ELECTRONICS ENGINEERING

SAINTGITS COLLEGE OF ENGINEERING

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KOTTAYAM

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CERTIFICATE

This is to certify that the report entitled **Ai-Assisted Wearable Mobility Aid For Visually Impaired** submitted by Afna A (MGP23ECE006) , Aswin S (MGP23ECE018) , Ebin Soyan Thomas (MGP23ECE020) , Juliya Augustine (MGP23ECE033) to SAINTGITS College of Engineering (Autonomous),affiliated to the APJ Abdul Kalam Technological University, Thiruvananthapuram, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Electronics and Computer Engineering, is a Bonafide record of project work done by them under my supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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DECLARATION

We undersigned hereby declare that the Mini Project report titled "**Ai-Assisted Wearable Mobility Aid For Visually Impaired**" submitted for partial fulfillment of the requirements for the award of the degree of Bachelor of Technology of Saintgits College of Engineering (Autonomous) affiliated to the APJ Abdul Kalam Technological University, Thiruvananthapuram is a Bonafide work carried out by us under the guidance of Er. Anish M George, Assistant Professor, Department of Electronics Engineering. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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ABSTRACT

This project presents the **Design and Implementation of an AI-Assisted Wearable Mobility Aid for Visually Impaired Individuals**. In everyday environments such as roads, public spaces, and unfamiliar surroundings, visually impaired individuals face significant challenges in safe and independent navigation. Traditional assistive tools like white canes provide limited information and are unable to detect dynamic or overhead obstacles effectively. To address these limitations, the proposed system introduces a cost-effective, wearable solution that enhances environmental awareness using modern embedded and artificial intelligence technologies.

The system utilizes a camera module to capture real-time visual data and estimate the distance of nearby obstacles using bounding box analysis. These inputs are processed using computer vision and lightweight machine learning techniques implemented on an embedded platform **(Raspberry Pi)**, enabling real-time obstacle detection and decision-making. The relative size and position of detected objects in the image frame are used to classify obstacles as near, medium, or far, providing an effective vision-based distance estimation approach. The processed information is conveyed to the user through an audio interface, providing timely alerts and navigation guidance without obstructing environmental sounds.

The proposed solution emphasizes portability, usability, and affordability while maintaining effective performance in real-world scenarios. The results demonstrate that the system improves navigation safety and user independence. This project highlights the practical application of embedded AI and assistive technology in enhancing the quality of life for visually impaired individuals and provides a scalable foundation for future developments such as advanced navigation and emergency assistance features.

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CHAPTER 1

INTRODUCTION

This project develops an AI-assisted wearable mobility aid using a camera-based system to detect obstacles in real time and estimate their distance using image-based bounding box analysis. It provides hands-free audio guidance to help visually impaired users navigate safely.

1.1 General

Ensuring safe and independent mobility for visually impaired individuals remains a major challenge, especially in dynamic environments like roads and crowded areas. Traditional aids such as white canes provide limited feedback and cannot detect overhead obstacles or contextual surroundings.

Although several electronic assistive systems have been developed, many of them are either expensive, bulky, or complex to use, limiting their accessibility and practical adoption. Furthermore, most existing solutions lack real-time intelligent perception and seamless user interaction, which are essential for enhancing navigation safety and user confidence.

This project develops an AI-assisted wearable mobility aid using camera-based vision techniques for real-time obstacle detection and distance estimation. It provides intuitive audio guidance to help visually impaired users navigate safely and independently.

The proposed solution is designed to be cost-effective, portable, and user-friendly while ensuring reliable real-time performance. It demonstrates a practical approach to enhancing mobility assistance using embedded AI, improving independence and quality of life for visually impaired individuals.

1.2 Objectives

The objectives define the key goals of developing an AI-based wearable system for safe and independent navigation.The main objectives are as follows:

Design of the System: To design and develop an AI-assisted wearable mobility aid integrated into a cap that enables safe and independent navigation for visually impaired users.

Real-Time Obstacle Detection: To detect obstacles in the user's path in real time using computer vision techniques and vision-based distance estimation.

Vision-Based Distance Estimation: To estimate the distance of detected objects using bounding box analysis by interpreting the relative size and position of objects in the image frame, enabling classification into near, medium, and far distances.

Embedded AI Implementation: To implement lightweight machine learning models on an embedded platform (Raspberry Pi) for efficient real-time object detection and decision-making.

Audio-Based Guidance: To provide clear and intuitive audio alerts and navigation instructions through a hands-free interface without blocking environmental sounds.

Voice Interaction: To enable basic voice command functionality for system control such as start, stop, and repeat instructions.

Wearable and Ergonomic Design: To design a lightweight, comfortable, and well-balanced cap structure that ensures proper placement of components without causing user discomfort.

Cost-Effective Solution: To build an affordable assistive device using low-cost and easily available hardware components.

Scalability and Future Enhancement: To design the system in a way that allows future integration of advanced features such as GPS navigation.

1.3 Scope of the Project

This project focuses on the design, development, and testing of an AI-assisted wearable mobility aid that enhances safe and independent navigation using computer vision and embedded systems. The key areas covered in this project are described below:

Application Scenario:

The proposed system is intended for use in real-world environments such as roads, public transportation areas, educational campuses, workplaces, and crowded public spaces. It assists visually impaired users in detecting obstacles and navigating safely in both indoor and outdoor conditions. The system is particularly useful in dynamic environments where traditional assistive tools provide limited support.

Technology Domain:

This project falls under the domains of embedded systems, artificial intelligence, computer vision, and assistive technology. It utilizes image processing techniques, machine learning models, and data acquisition from vision-based inputs to achieve real-time environmental perception and decision-making.

System Design and Architecture:

The system is designed as a wearable cap integrating a camera module, Raspberry Pi processing unit, audio output interface, and power supply. The architecture follows an edge-based processing approach where all computations are performed locally on the embedded system, ensuring minimal latency and independence from cloud services.

Obstacle Detection and Perception:

The project focuses on detecting obstacles in the user's path using computer vision and vision-based distance estimation techniques. The camera captures real-time video data, which is processed using lightweight AI models to identify objects. The relative size and position of detected objects in the image frame are used to estimate their distance and classify them as near, medium, or far, improving the system's perception capability.

Vision-Based Distance Estimation:

A key aspect of the system is estimating the distance of detected objects using bounding box analysis. By analyzing the dimensions and position of bounding boxes in the captured frames, the system infers obstacle proximity without the need for additional distance sensors, thereby maintaining system simplicity while ensuring effective performance.

Audio Guidance System:

The processed information is delivered to the user through an audio interface in the form of alerts and navigation cues. The system ensures hands-free operation and is designed to provide clear and timely feedback without blocking environmental sounds, thereby maintaining user awareness.

Wearable Design and Ergonomics:

The system is implemented as a cap-based wearable device to ensure comfort and ease of use. Component placement is optimized to maintain weight balance, with the camera module positioned at the front and processing and power units distributed on either side. The design prioritizes portability and user convenience.

Power Management:

The system is powered using a portable power bank, ensuring mobility and continuous operation. Efficient power management techniques are considered to optimize battery life while supporting real-time processing requirements.

Cost and Accessibility:

The project emphasizes the use of low-cost and readily available components to ensure affordability. The goal is to develop an assistive solution that is accessible to a wider population without compromising essential functionality.

System Performance and Testing:

The system is evaluated based on parameters such as obstacle detection accuracy, response time, reliability, and usability. Testing is conducted in controlled environments to analyze performance under different conditions, including varying lighting and obstacle scenarios.

Future Scope:

The system is designed with scalability in mind, allowing future enhancements such as GPS-based navigation, emergency alert systems, improved vision-based distance estimation techniques, advanced object classification models, and integration with mobile applications. Further miniaturization and optimization can improve usability and enable real-world deployment.

Summary:

Overall, the scope of this project covers the complete development cycle of an AI-based wearable assistive system, focusing on real-time obstacle detection, vision-based distance estimation, user guidance, and practical usability. It provides a strong foundation for further research and development in assistive technologies aimed at improving the independence and quality of life of visually impaired individuals.

CHAPTER 2

LITERATURE REVIEW

A comprehensive review of existing literature was carried out to understand the current state of assistive technologies for visually impaired individuals and the various techniques used in real-time object detection and navigation systems. The following table summarizes the key papers reviewed:

Table 2.1: Consolidation of Literature Review

Sl. No.	**Title**	**Authors & Publication**	**Techniques Used and Observations**
1	Wearable Navigation Device for Visually Impaired People	Z. Chen, X. Liu, M. Kojima, Q. Huang, T. Arai, Sensors, 2021.	**Technique Used:** Semantic Visual SLAM with RGB-D camera and deep learning. **Observation:** Provides accurate real-time environment mapping and object detection. **Objective:** To assist visually impaired users with real-time navigation and scene understanding.
2	Intelligent Navigation Assistance for Visually Impaired People	Y. Said, M. Atri, M. A. Albahar, A. B. Atitallah, Y. A. Alsariera, Sensors, 2023.	**Technique Used:** Deep learning with Neural Architecture Search (NAS). **Observation:** Improves obstacle detection accuracy and adaptability. **Objective:** To develop an intelligent navigation system using optimized AI models.

Sl. No.	Title	Authors & Publication	Techniques Used and Observations
3	Wearable Obstacle Avoidance Electronic Travel Aids for Blind: A Survey	D. Dakopoulos, N. G. Bourbakis, IEEE Transactions, 2010.	**Technique Used:** Survey of ultrasonic, camera, and sensor-based systems. **Observation:** Multi-sensor systems improve navigation reliability. **Objective:** To analyze and compare existing assistive navigation technologies.
4	Drishti: An Integrated Indoor/Outdoor Blind Navigation System	L. Ran, S. Helal, S. Moore, IEEE Pervasive Computing, 2004.	**Technique Used:** GPS + GIS + computer vision integration. **Observation:** Enables seamless indoor and outdoor navigation. **Objective:** To develop a unified navigation system for blind users.
5	Wearable Assistive Devices for Visually Impaired: A State-of-the-Art Survey	R. Tapu, B. Mocanu, T. Zaharia, Pattern Recognition Letters, 2014.	**Technique Used:** Review of computer vision and wearable assistive systems. **Observation:** Vision-based systems offer better environmental understanding. **Objective:** To summarize advancements in wearable assistive technologies.
6	Smart Real-Time Wearable Navigation Support System for Blind and Visually Impaired People	Y. Bouteraa, I. B. Abdallah, N. Derbel, Journal of Intelligent Systems, 2022.	**Technique Used:** Fuzzy logic with sensor fusion and embedded systems. **Observation:** Enhances decision-making and reduces navigation errors. **Objective:** To design a real-time assistive wearable system using intelligent control.

Sl. No.	Title	Authors & Publication	Techniques Used and Observations
7	Obstacle Detection and Navigation System for Visually Impaired Using Raspberry Pi	K. Patil, P. Kulkarni, IJERT, 2022.	**Technique Used:** Raspberry Pi with ultrasonic sensors and camera-based detection. **Observation:** Provides low-cost and effective obstacle detection. **Objective:** To develop an affordable assistive navigation system.
8	Real-Time Object Detection System for Visually Impaired Using YOLO	S. Redmon et al., IEEE Conference (YOLO-based studies), 2016.	**Technique Used:** YOLO (You Only Look Once) deep learning object detection. **Observation:** Achieves high-speed real-time detection with good accuracy. **Objective:** To enable fast and efficient object recognition for assistive applications.

2.1 Gaps in Literature

A review of existing assistive navigation systems for visually impaired individuals shows that many solutions rely on vision-based techniques alone, which can limit overall system performance when distance estimation is not effectively addressed. While vision-based systems offer object recognition and environmental understanding, many lack reliable and efficient methods for estimating obstacle distance in real time, especially on low-power embedded platforms. Additionally, some systems suffer from latency and reduced performance in varying lighting conditions.

Several existing solutions also depend on cloud processing, leading to delays and reduced reliability in real-time applications. Many systems are bulky, expensive, or not designed for wearable comfort, reducing their practical usability. Furthermore, limited implementation of efficient edge-based processing and optimized vision algorithms results in lower performance and slower decision-making.

These gaps highlight the need for a compact, cost-effective, and wearable system that utilizes computer vision with efficient bounding box–based distance estimation and edge-based processing to provide reliable real-time obstacle detection and intuitive user guidance, which is the focus of this project.

CHAPTER 3

COMPONENTS AND WORKING METHODOLOGY

3.1 Hardware Components

The following hardware components are used in the development of the proposed wearable assistive system.

Table 3.1: Hardware Components (Part 1)

Sl. No.	**Component**	**Function**
1	Raspberry Pi 4 Model B	Acts as the main processing unit of the system. It performs real-time image processing, executes computer vision algorithms using OpenCV, and runs AI models for object detection and decision-making.
2	Camera Module (Raspberry Pi Camera/Usb Camera)	Captures real-time video of the surrounding environment. Mounted at the front curve of the cap to align with the user's field of view for accurate obstacle detection.
3	Power Bank (5V, 4000–5000 mAh)	Provides portable power supply to the Raspberry Pi and connected components, ensuring continuous system operation.
4	Microphone Module	Captures user voice commands such as start, stop, and repeat. Enables hands-free interaction with the system.

Table 3.2: Hardware Components (Part 2)

Sl. No.	**Component**	**Function**
5	Earphones / Bone Conduction Headset	Provides audio output in the form of alerts and navigation instructions while allowing environmental sounds to be heard.
6	Raspberry Pi Protective Case	Protects the Raspberry Pi from dust, heat, and physical damage while enabling secure mounting on the cap.
7	Jumper Wires and Ribbon Cable	Used for electrical connections between Raspberry Pi and camera module, ensuring reliable data transmission.
8	Wearable Cap (Mounting Platform)	Acts as the base structure for mounting all components. Designed for comfort and balanced weight distribution.

Summary: The hardware components selected for this system collectively enable real-time obstacle detection, processing, and user guidance. The integration of the camera module, embedded processing unit, and audio output ensures efficient performance while maintaining portability and user comfort. The design emphasizes cost-effectiveness, reliability, and balanced weight distribution, making the system suitable for practical use by visually impaired individuals.

3.2 Software Components

The software components used in this project play a crucial role in enabling real-time processing, obstacle detection, and user interaction. The system is designed using a combination of programming languages, libraries, and frameworks optimized for embedded AI applications.

Operating System (Raspberry Pi OS): A Linux-based operating system used to manage hardware resources and provide a stable platform for running all software applications and libraries required for the system.

Programming Language (Python): Python is used as the primary programming language due to its simplicity, flexibility, and extensive support for AI and computer vision libraries. It is used for implementing system logic and data processing.

OpenCV (Computer Vision Library): OpenCV is used for image processing and real-time video analysis. It enables frame capture, preprocessing, object detection, and feature extraction from the camera input.

TensorFlow Lite (Machine Learning Framework): TensorFlow Lite is used to run lightweight deep learning models on the Raspberry Pi. It enables efficient real-time object detection and classification with low latency and reduced computational load.

Data Processing and Decision Logic: Custom Python scripts are used to process outputs from AI models and make decisions regarding alerts and navigation instructions.

Summary: The software components of the proposed system enable real-time processing, perception, and user interaction through an efficient embedded AI framework. The combination of Raspberry Pi OS and Python provides a flexible platform for implementing system logic, while OpenCV handles image processing and video analysis. TensorFlow Lite supports lightweight deep learning models for real-time object detection, ensuring low latency and efficient performance on embedded hardware.

Additionally, text-to-speech modules convert system outputs into intuitive audio guidance. Optional speech recognition enhances user interaction through voice commands. Overall, the software architecture ensures reliability, scalability, and responsiveness, making it suitable for real-world assistive applications and future enhancements.

3.3 Hardware Design Proposal

The hardware design represents the physical implementation of the AI-assisted wearable mobility aid, focusing on compactness, comfort, and efficient operation, as shown in Fig. 3.1.

The system is implemented as a wearable cap. The USB camera is mounted at the front to capture real-time visual data aligned with the user's field of view, while the Raspberry Pi 4 Model B and power supply are positioned to ensure balanced weight distribution.

Audio output is provided through earphones or a bone-conduction headset, and a microphone module enables voice-based interaction. The overall design ensures secure mounting, portability, and reliable real-time performance for practical assistive use.



Figure 3.1: Hardware Design of AI-Assisted Wearable Mobility Aid

3.4 Working Methodology

The proposed system operates through a well-defined pipeline that integrates sensing, processing, and feedback mechanisms to assist visually impaired users in real-time navigation. The methodology is aligned with the system block diagram and is divided into multiple stages as follows:

1. **Data Acquisition:** The system begins by capturing real-time environmental data using a camera module. The camera continuously records video frames from the front view of the cap.
2. **Pre-processing:** Captured video frames are preprocessed using OpenCV techniques such as resizing, noise reduction, and frame normalization to prepare the data for efficient analysis.
3. **Obstacle Detection using Computer Vision:** The processed frames are passed through a lightweight AI model implemented using TensorFlow Lite. The model detects objects and obstacles present in the user's path and classifies them accordingly.
4. **Distance Estimation:** The system estimates the distance of detected objects using bounding box analysis. The relative size and position of bounding boxes in the image frame are used to determine whether an object is near, at a medium distance, or far from the user.
5. **Decision Making:** Based on the detected object and its estimated distance, the system determines whether an obstacle is present and decides the appropriate response, such as issuing a warning or guiding the user.
6. **Audio Output Generation:** The decision output is converted into speech using text-to-speech modules. Audio alerts such as "Obstacle ahead" or directional guidance are generated.
7. **User Feedback:** The generated audio is delivered through earphones or a bone-conduction speaker, allowing the user to receive real-time guidance without blocking environmental sounds.
8. **Continuous Monitoring:** The system operates continuously in a loop, updating visual data and processing frames in real time to ensure smooth and uninterrupted navigation.
9. **Power Management and System Control:** The system is powered using a portable power bank, ensuring mobility. Optional voice commands allow users to control system operations.

Summary: The working methodology of the system is based on a continuous pipeline that integrates sensing, processing, and feedback to assist visually impaired users in navigation. Environmental data is captured using a camera, providing visual information about the surroundings. This data is processed using computer vision and lightweight AI models on the Raspberry Pi to detect and classify obstacles in real time.

Distance estimation is performed using bounding box analysis, where the relative size and position of detected objects are used to determine their proximity. The decision-making module determines appropriate alerts based on obstacle presence and estimated distance. These alerts are delivered through an audio interface, enabling hands-free guidance without blocking environmental awareness. The system ensures real-time operation, portability, and user comfort, making it an effective and practical assistive solution.

3.5 Block Diagram

The block diagram represents the overall architecture of the proposed AI-assisted wearable mobility aid system, as shown in Fig. 3.2. It illustrates the flow of data from Camera Input through the processing unit to the output interface.

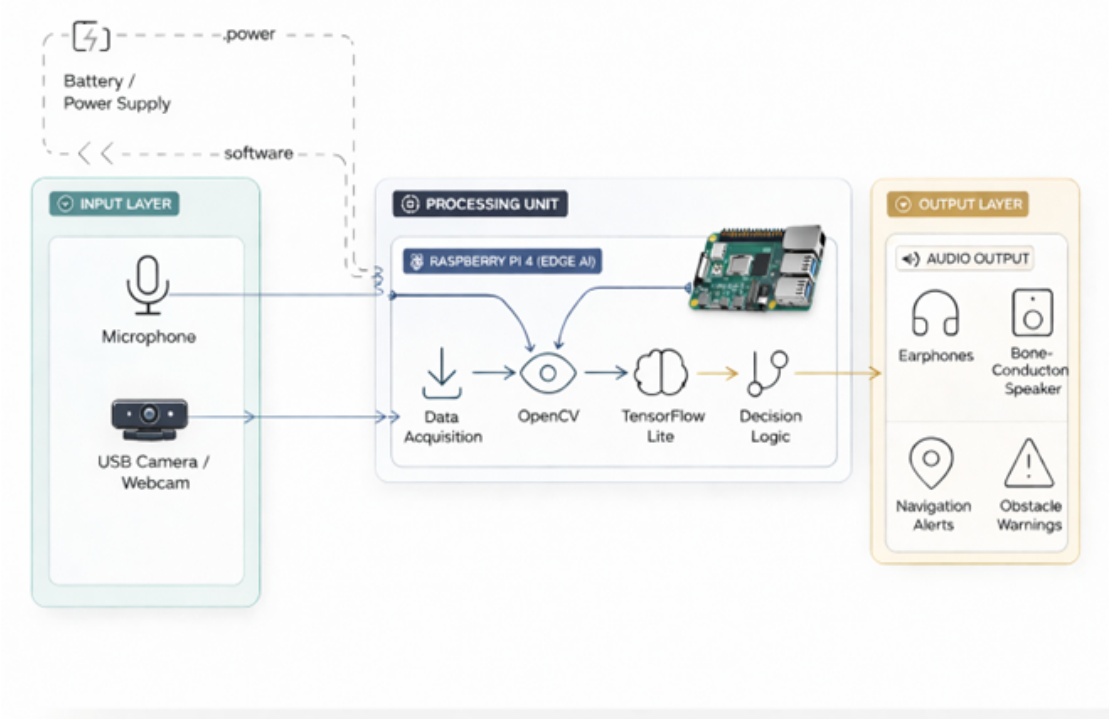


Figure 3.2: Block Diagram of AI-Assisted Wearable Mobility Aid for Visually Impaired

3.6 Circuit Diagram

The circuit diagram shows the electrical connections between all hardware components of the system, as shown in Fig 3.3.

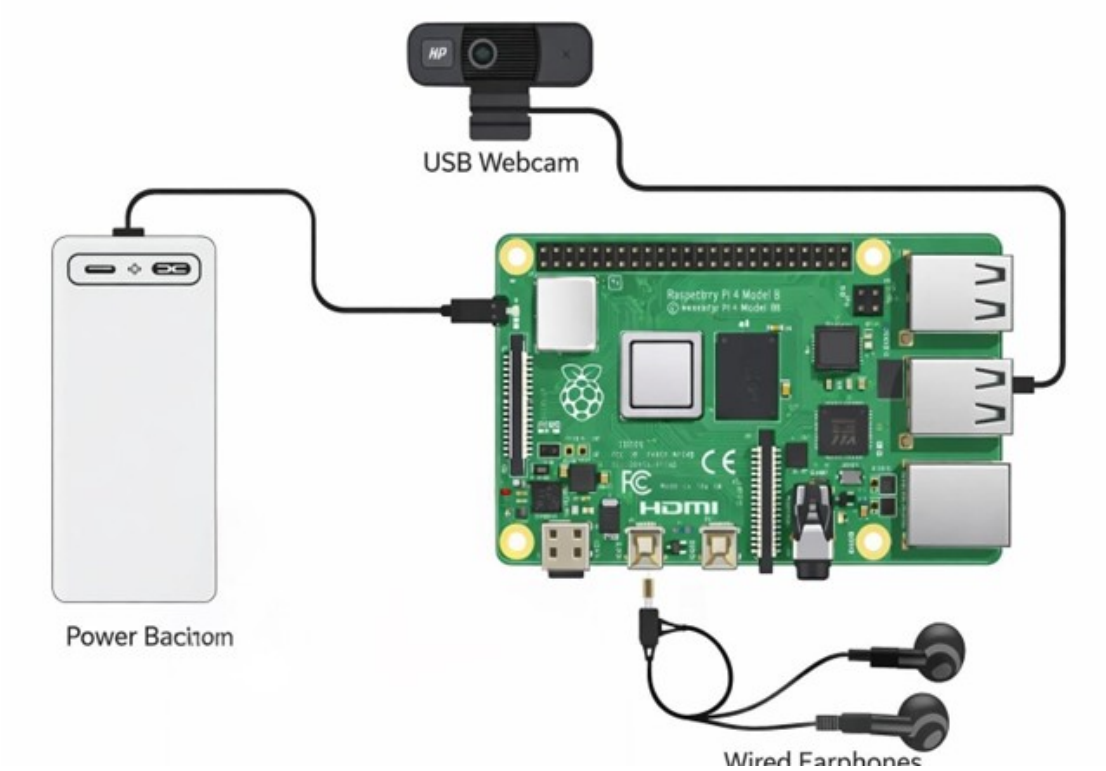


Figure 3.3: Circuit Diagram of AI-Assisted Wearable Mobility Aid for Visually Impaired

The circuit diagram illustrates the interconnection of the **Raspberry Pi 4 Model B, USB camera, microphone module, earphones/bone conduction headset**, and the power management module. It shows how visual and audio data are processed for real-time obstacle detection, voice assistance, and navigation support in the AI-assisted wearable mobility aid for visually impaired individuals.

CHAPTER 4

WORK PROGRESS AND RESULTS

4.1 Hardware Prototype

The hardware prototype of the proposed system was developed to implement real-time assistance and navigation support for visually impaired individuals using embedded components. The core of the system is the **Raspberry Pi 4 Model B**, which performs image processing, object detection, and voice assistance operations. The **USB camera** is used to capture the surrounding environment, while the **microphone module** enables voice input and interaction with the system. The processed information and navigation instructions are delivered through the **earphones/bone conduction headset**.

The components were interconnected to ensure efficient and reliable data communication between all modules. Proper connections and power supply were established to maintain stable operation of the system. The prototype was assembled in a compact and wearable setup, making it suitable for practical daily use.

Testing of the hardware prototype confirmed successful integration of all components, with proper image capture, voice interaction, and audio output functionality. The system demonstrated reliable performance in detecting obstacles and assisting user navigation, validating the effectiveness of the hardware design.

4.2 Software Development Phase

The software development phase focused on implementing real-time object detection, voice assistance, and navigation support on the **Raspberry Pi 4 Model B**. The system software was developed using **Python** along with libraries such as **OpenCV** and AI-based object detection frameworks. Initially, the **USB camera** and **microphone module** were configured to enable image capture and voice interaction.

In the next stage, **image acquisition and processing** were established by continuously capturing the surrounding environment through the camera. The captured frames were processed in real time to identify obstacles and important objects. Additionally, bounding box analysis was implemented to estimate the distance of detected objects based on their relative size and position in the image frame. Proper synchronization between image processing and audio feedback was maintained to ensure smooth system performance and user interaction.

Further development involved implementing **AI-based object detection and voice assistance algorithms** to provide navigation guidance and environmental awareness. The system was optimized to maintain **fast response time** and reliable performance during real-time operation. Several rounds of testing and tuning were carried out to improve detection accuracy and ensure stable operation under different conditions.

The results indicate that the system successfully performs real-time object detection, vision-based distance estimation, and voice-based assistance with reliable performance. The audio guidance provided clear navigation support and obstacle awareness, confirming the effectiveness of the implemented software in assisting visually impaired individuals.

4.3 Results and Testing

The developed AI-assisted wearable mobility aid was tested under different environmental conditions to evaluate its performance in assisting visually impaired individuals with navigation and obstacle awareness. Various test conditions involving obstacle detection and real-time navigation assistance were considered during system evaluation. The key results observed are summarised below. The output detected images are shown in Fig. 4.1, Fig. 4.2, and Fig. 4.3.

- The system successfully captured real-time visual data using the **USB camera** for environment monitoring and object detection.
- The **Raspberry Pi 4 Model B** effectively processed the captured images and executed **AI-based object detection and navigation algorithms** in real time.
- The implemented system accurately detected nearby obstacles and important objects, providing reliable assistance to the user.
- The system maintained **fast response time during real-time operation**, ensuring smooth and uninterrupted navigation support.
- The generated audio guidance through the **earphones/bone conduction headset** provided clear instructions and improved user awareness of the surroundings.
- The **microphone module** enabled effective voice interaction and user communication with the system.
- The overall system performance remained stable under different environmental conditions, validating the effectiveness of the AI-assisted wearable mobility aid for visually impaired individuals.

4.4 Output Results

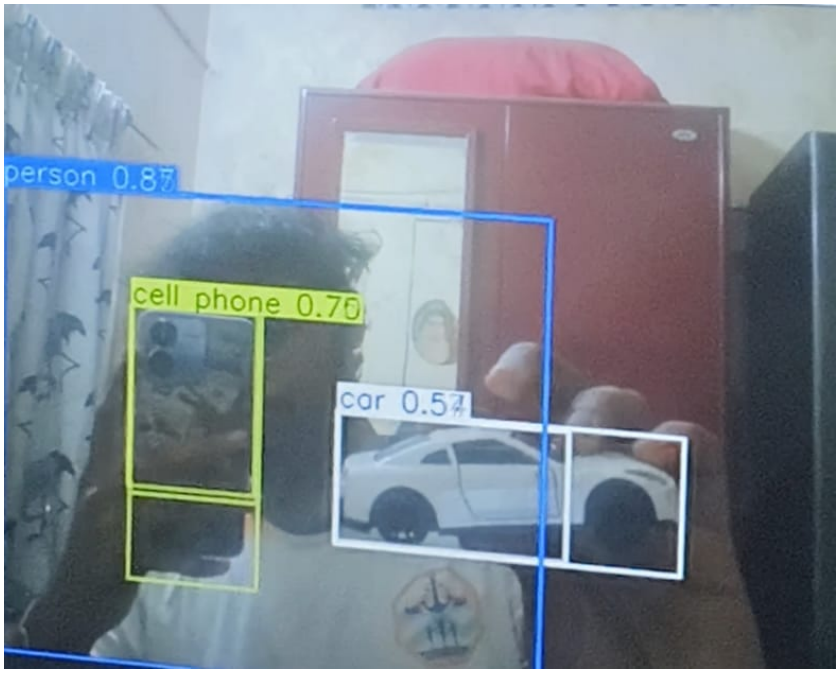


Figure 4.1: Real-Time Multi-Object Detection Output



Figure 4.2: Mobile Phone Detection Output



Figure 4.3: Chair Detection Output

4.5 Cost Estimation

The cost estimation for all hardware components used in the prototype is detailed in Table 4.1.

Table 4.1: Cost Estimation of the Prototype

Sl. No.	**Component**	**Estimated Cost (Rs.)**
1	Raspberry Pi 4 (4GB)	7000
2	Case with Cooling Fan	670
3	USB Webcam	800
4	Earphones / Bone Conduction Headset	450
5	Power Bank	1500
Total Estimated Cost		**Rs. 10,420**

The hardware prototype was developed at a total estimated cost of approximately Rs. 10,420, demonstrating that an effective AI-assisted wearable mobility aid can be built at a reasonable cost. The use of components such as the **Raspberry Pi 4, USB webcam, microphone module, and earphones/bone conduction headset** ensures that the system delivers reliable performance for real-time object detection and navigation assistance. This makes the solution suitable for practical everyday use by visually impaired individuals in environments such as homes, public spaces, and workplaces.

CHAPTER 5

CHALLENGES

The development of the AI-assisted wearable mobility aid involved several technical and practical challenges. The major challenges encountered are described below:

1. **Real-Time Processing:** Achieving real-time object detection and navigation assistance was challenging due to the high computational requirements of image processing and AI algorithms. Proper optimization was required to ensure smooth and fast system response.
2. **Accurate Object Detection:** Detecting obstacles and important objects accurately in different environments was difficult because of variations in lighting conditions, object size, and background complexity.
3. **Voice Interaction and Audio Feedback:** Ensuring clear voice interaction and accurate audio guidance was challenging, especially in noisy surroundings. Proper tuning of the microphone and audio output system was necessary for effective communication.
4. **Maintaining System Performance:** Running continuous image processing while maintaining stable performance and low response time required efficient resource management and software optimization.
5. **Hardware Integration and Communication:** Integrating the Raspberry Pi, USB camera, microphone module, and audio output devices into a compact wearable setup was challenging. Stable connections and proper power management were essential for reliable operation of the system.

CHAPTER 6

MEMBER ROLES

Table 6.1: Member Roles

Member	**Name (Roll No.)**	**Responsibilities**
Member 1	Afna A (MGP23ECE006)	Research on components and their specifications, implementation methods, and literature review.
Member 2	Aswin S (MGP23ECE018)	Coding and study of software components, hardware design, and system integration.
Member 3	Ebin Soyan Thomas (MGP23ECE020)	Dataset collection, coding of software components, and literature review.
Member 4	Juliya Augustine (MGP23ECE033)	Dataset collection, system setup and installation, documentation, and literature review.

CHAPTER 7

CONCLUSION

The project successfully achieved the design and implementation of an AI-assisted wearable mobility aid for visually impaired individuals. The system was able to capture real-time environmental data and process it using AI-based object detection and navigation techniques to improve user safety and mobility. The wearable design makes the system practical and convenient for everyday use.

The integration of hardware components such as the **Raspberry Pi 4 Model B, USB camera, microphone module, and earphones/bone conduction headset** enabled efficient real-time image processing and voice assistance. The system demonstrated stable performance in capturing visual information, detecting obstacles, and providing audio guidance to the user. Proper integration of hardware and software ensured smooth communication between all components.

A key feature of the system is the implementation of vision-based distance estimation using bounding box analysis. Instead of relying on additional distance sensors, the system determines the proximity of detected objects based on their relative size and position in the image frame. This approach simplifies the hardware design while maintaining effective obstacle awareness and real-time performance.

The results obtained from the prototype confirm that the system is capable of providing reliable navigation support and obstacle awareness for visually impaired individuals. The solution proved to be cost-effective, portable, and suitable for use in environments such as homes, public spaces, and workplaces.

Overall, this project highlights the practical application of artificial intelligence and embedded systems in assistive technology. It provides an effective approach to improving independent mobility and user safety, while also serving as a foundation for further advancements in AI-based wearable assistance systems.

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